



DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING

Devarakagalahalli, Harohalli, Kanakapura Road, Bengaluru South, Karnataka - 562 112

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

Project – SYNOPSIS

1. Project Title:

A Generative AI Framework for Automatic Comic Strip Generation

2. Team Details:

Sl. No	Name of Student	USN	Email ID	Contact No.
1.	Akhila Rao D	ENG22AM0002	eng22am0002@dsu.edu.in	9481389538
2.	GaanaShree S	ENG22AM0014	eng22am0014@dsu.edu.in	9663984811
3.	Nitya P Shetty	ENG22AM0037	eng22am0037@dsu.edu.in	9036603724

3. Domain Specifications:

Generative AI and Creativity

4. Sustainability Development Goals

1. SDG 4 – Quality Education

The project supports creative and inclusive learning by enabling teachers and students to generate visual storytelling content instantly. This empowers learners with diverse abilities, enhances engagement, and promotes innovative teaching methods.

2. SDG 9 – Industry, Innovation, and Infrastructure

By integrating LLMs, diffusion models, and user-friendly interfaces like Gradio, the project demonstrates technological innovation. It encourages the adoption of advanced AI tools and supports digital infrastructure development for creative industries.

3. SDG 17 – Partnerships for the Goals

The project demonstrates collaboration between open-source communities, cloud AI services, and educational platforms. It highlights the value of partnerships in advancing AI-driven creativity and knowledge sharing.

5. Problem Statement:

- Traditional comic creation requires artistic skills, manual drawing, and significant time investment, making it inaccessible to many learners and creators.
- Existing tools for comic generation are either complex, costly, or lack automation, limiting creative expression for beginners or non-artists.
- There is a growing need for AI-assisted tools that convert simple text prompts into high-quality visual narratives efficiently and consistently.

6. Objectives of the Project:

- To develop an AI-driven system capable of generating comic panels directly from text prompts using lightweight diffusion models.
- To automate the transformation of user-provided descriptions into sequential visual stories with stylistic consistency.
- To provide a simple Gradio-based interface that allows users to input text, generate panels, and download results easily.
- To reduce the time and effort required for comic creation by leveraging fast and resource-efficient AI models.
- To support creative learning, storytelling, and visualization across educational and content-creation domains.

7. Project Scope:

The scope of this project encompasses the design and development of an AI-powered comic generation system that converts user-provided text prompts into visually coherent comic panels. It includes implementing a Gradio-based user interface, integrating a lightweight large language model for structured panel descriptions, and using a diffusion model to generate high-quality images. The project covers text-to-image generation, panel composition, text rendering within panels, and export of the final comic strip. Additionally, it includes performance optimization, minimal hardware support, and ensuring user-friendly interaction. However, the scope does not extend to advanced animation, multi-page comic books, or real-time collaborative editing; instead, it focuses on delivering a fast, accessible, and automated pipeline for single-strip comic creation.

8. Source of Data Collection:

● Kaggle Datasets

1. **FiveThirtyEight Comic Characters Dataset:** <https://www.kaggle.com/datasets/fivethirtyeight/fivethirtyeight-comic-characters-dataset>
2. **Superhero Character and Powers:** <https://www.kaggle.com/datasets/baraaaid/superherodb>
3. **The Marvel Universe Social Network:** <https://www.kaggle.com/datasets/csanhueza/the-marvel-universe-social-network>
4. **DC Comic Books Dataset:** <https://www.kaggle.com/datasets/deepcontractor/dc-comic-books-dataset>

● GitHub Repositories (Code & Data)

1. **macrae/comics-net (Comic Cover Image/Metadata):** <https://github.com/macrae/comics-net>
2. **tashapiro/superhero-comics (Superhero API Data):** <https://github.com/tashapiro/superhero-comics>

9. Expected Outcomes:

The expected outcome of this project is a fully functional AI-driven system capable of generating coherent and visually appealing comic panels from simple text prompts. Users will be able to input a storyline and receive a structured, multi-panel comic strip with consistent characters, appropriate scenes, and clearly rendered dialogue. The system is expected to produce results within a few seconds, ensuring fast and interactive feedback. Additionally, it will reduce the effort required for manual comic creation, making storytelling more accessible for students, educators, content creators, and hobbyists. Overall, the project will demonstrate how generative AI can automate creative processes while maintaining quality, usability, and efficiency.

10. Tools, Technologies & Platforms to be Used:

Python – Core programming language used to develop the entire comic generation pipeline.

Gradio – Provides a simple web interface for users to input prompts and view generated comics.

OpenAI LLM – Generates structured panel descriptions and dialogues based on user prompts.

Stability AI (Stable Diffusion) – Creates comic-style images from the LLM-generated descriptions.

Pillow (PIL) – Handles text rendering and composes individual images into a final comic strip.

JSON / Pydantic – Validates structured outputs received from the language model.

Hugging Face Spaces – Hosts and deploys the full application for public access.

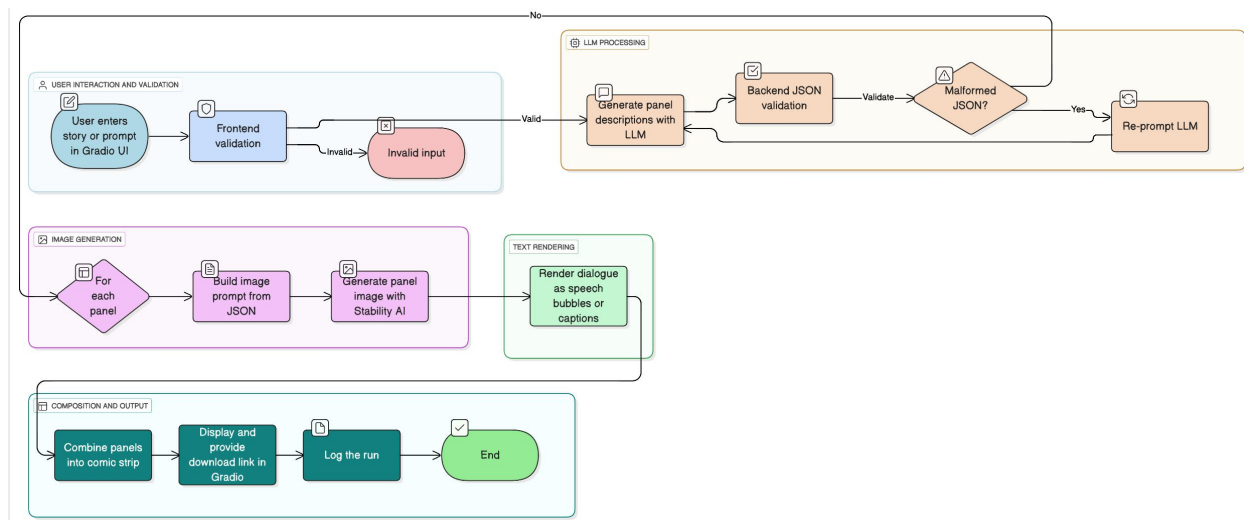
GitHub – Stores source code and project documentation with version control.

11. Methodology/Approach:

The project begins by designing a user-friendly interface using Gradio, where users can provide a short prompt, storyline, or scene description. This text input is processed by a lightweight Large Language Model (LLM), which is instructed through prompt engineering to generate a structured JSON output. The LLM breaks the storyline into meaningful comic panels, describing characters, background settings, emotions, and dialog for each scene. This structured, machine-readable format ensures consistency between panels and prevents hallucinations, enabling precise downstream processing.

Once the panel descriptions are generated, each panel's visual components are sent to a diffusion-based image generator (Stable Diffusion / Stability AI). The model generates images based on the detailed prompts while maintaining stylistic consistency across the strip. After generating the images, the system uses Python's Pillow library to render dialogues as text overlays, ensuring readable font placement and bubble alignment. Each panel is then standardized to a fixed size to maintain uniformity, and all panels are assembled vertically or horizontally depending on the selected layout.

The final stage involves combining the rendered panels into a complete comic strip and returning it to the user through the Gradio interface. Throughout the pipeline, error-handling mechanisms are applied—such as JSON validation, fallback prompt generation, and safe API checks—to ensure predictable results. The workflow is optimized for minimal hardware use, ensuring the application runs smoothly even with limited GPU resources. The end-to-end methodology enables fast, automated, and high-quality comic creation from any user prompt, demonstrating the practical integration of generative AI in creative storytelling.



Workflow

12. References :

1. Mohammed Mahmood Ahmed, Dr. Abdul Rahman2, ``AI Comic Strip Generator Using Natural Language Processing and Image Synthesis," INTERNATIONAL JOURNAL OF RECENT TRENDS IN MULTIDISCIPLINARY RESEARCH <https://www.doi.org/10.59256/ijrtmr.20250505012>
2. Xin Yang Dalian University of Technology, Zongliang Ma, Letian Yu, Ying Cao East China University of Science and Technology`Automatic Comic Generation with Stylistic Multi-page Layouts and Emotion-driven Text Balloon Generation" ResearchGate arXiv:2101.11111v1 [cs.CV]
3. Pramoda P Gunasekara Faculty of Computing, Sri Lanka Institute of Information Technology,Pawani Muthusala Perera Faculty of Computing, Sri Lanka Institute of Information Technology,`Generate Comic Strips Using AI" Vol. 1 No. 1 (2024): Proceedings of Conference on Transdisciplinary Research in Engineering
4. Yi-Chun Chen, Arnav Jhala, ``Collaborative Comic Generation: Integrating Visual Narrative Theories with AI Models for Enhanced Creativity"arXiv:2409.17263 [cs.AI] (or arXiv:2409.17263v1 [cs.AI] for this version) <https://doi.org/10.48550/arXiv.2409.17263>
5. Mohammed Rakibul Hassan, Madiahah Sheikh Abdul Aziz* Dept. of Information Systems, KICT, International Islamic University Malaysia, 53100, Kuala Lumpur, Malaysia, ``Artificial Intelligence Comic Strip (AICS) Generators: A Review of Subscription Models, Pricing, and User Satisfaction"International Journal on Perceptive and Cognitive Computing (IJPCC) <https://doi.org/10.31436/ijpcc.v11i1.553>
6. Krunoslav Bedi School of Civil Engineering, Art and Craft Čakovec Športska 1, Čakovec, Croatia, ``AI Comics as Art: Scientific Analysis of the Multimedia Content of AI Comics in Education"2023 46th MIPRO ICT and Electronics Convention (MIPRO) Doi:0.23919/MIPRO57284.2023.10159693